# MATH 3007 Optimization

The Chinese University of Hong Kong (Shenzhen), Spring 2023

**Instructor:**

Dr. Zizhuo Wang
Office: Daoyuan Building 407
Email: wangzizhuo@cuhk.edu.cn

**Meeting Times/Location:**

Wednesday and Friday 10:30 AM – 11:50 AM
Location: Administrative Building E206

**Office Hours:**

Wednesday 3:00 PM–4:30 PM
or by appointment

**Tutorial Sessions Hours:**

Time: Mon/Wed 6:00PM-6:50PM/7:00PM-7:50PM/8:00PM-8:50PM
Location: Teaching Building C-209
Instructors: See Schedule

**Reference Books:**

1. Introduction to Linear Optimization, by D. Bertsimas and J. Tsitsiklis
2. Convex Optimization, by S. Boyd and L. Vandenberghe (electronic copy available online)

**Course Website:**

https://bb.cuhk.edu.cn/ and find MATH3007. All course materials will be posted there.

**Teaching Assistant:**

Yicheng Liu
Email: yichengliu@link.cuhk.edu.cn
Office Hours: Tuesdays 7:00PM-9:00PM
Location: TBD

Wenjun Zheng
wenjunzheng@link.cuhk.edu.cn

**Course Summary:** This course focuses on the basics of optimization theories and applications. The first part of the course will be mainly about linear optimization, including how to formulate a linear program, the simplex method and the duality theory. Then we will study nonlinear optimization, focusing on the optimality conditions and some useful algorithms. Integer programming will also be briefly discussed in this course.

**Homework:** There will be about 9 homework assignments. You can consult with other students on how to do the homework problems, but all homework must be completed individually. Electronic submissions are strongly preferred.

**Programming:** In this course, we will introduce how to use computer program to solve optimization problems. We will mainly use MATLAB (with cvx package) and Python (with cvxpy package). You are expected to be familiar with at least one type of these two programming languages by the end of the course.

**Grading:** Grades will be based upon homework assignments, a midterm exam and a final exam. The exam schedule is:

| | |
|---|---|
| Midterm: | Saturday, Mar 25th, 10-11:30am |
| Final Exam: | TBD |

If you miss an exam without prior permission, then you will be given a 0 on that exam (and will be reported to the registry). If you know you have a conflict in exam time, please Email me as early as possible.

Course grades will be broken down as follows:

| | |
|---|---|
| Midterm | 40% |
| Final Exam | 40% |
| Homework | 20% |

**Course Outline:** The following is a rough outline of the course topics:

| Week | Topics |
|---|---|
| 1 | Introduction to optimization |
| 2-4 | Basics about linear optimization, the simplex algorithm |
| 4-5 | Duality theories of linear programs |
| 6 | Dual simplex, sensitivity analysis and more on LP |
| 7 | Review and midterm |
| 8-10 | Introduction to nonlinear optimization, optimality conditions |
| 11-12 | Algorithms for nonlinear optimization |
| 13-15 | Integer programs, introductions and basic algorithms, and final review |